

Spatial Statistics - Homework 5.3

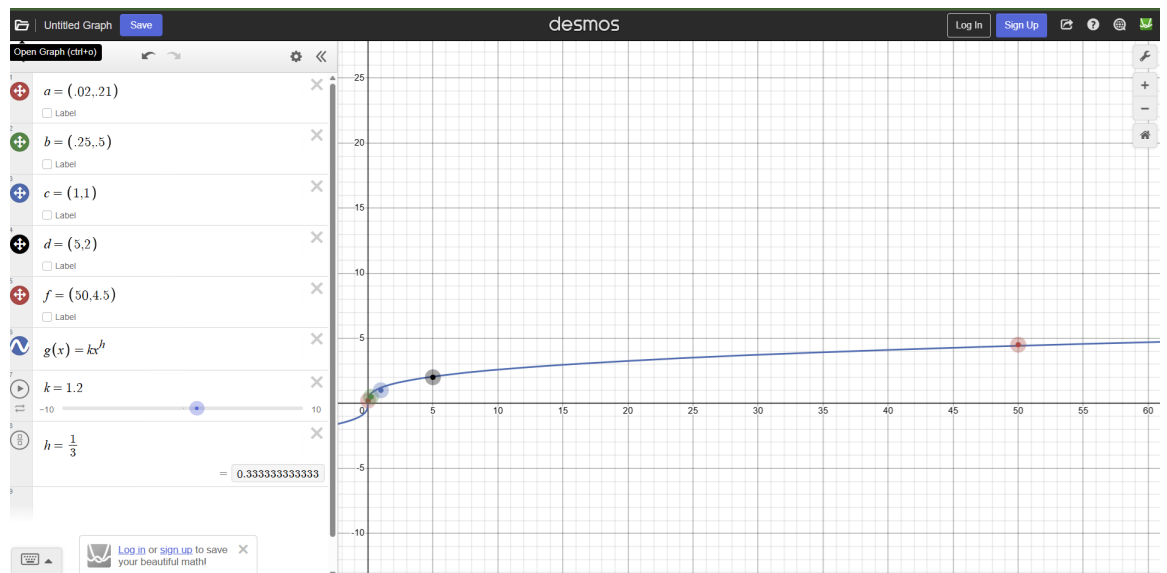
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1. 6.5

Solution: (a) I believe that I am not super amazing at viewing log-log plots, but the slope of the line seems to be $\frac{1}{3}$,

When plotting it in desmos, I got the following graph:



I also tried combinations of log slopes with the points and the values of the slopes fluctuated a lot between .3 and .5, but I think the best fit is $\frac{1}{3}$.

(b) The parametrization of this model should be given by the first odd power term in the Taylor expansion of the function. The Taylor expansion of $\theta_1 e^{-r/\theta_2} (1 + r/\theta_2)$

is given by:

$$\begin{aligned}\theta_1 e^{-r/\theta_2} (1 + r/\theta_2) &= \theta_1 \left(1 - \frac{r}{\theta_2} + \frac{r^2}{2\theta_2^2} - \frac{r^3}{6\theta_2^3} + \frac{r^4}{24\theta_2^4} \cdots \right) \left(1 + \frac{r}{\theta_2} \right) \\ &= \theta_1 \left(1 - \frac{r^2}{2\theta_2^2} + \frac{r^3}{3\theta_2^3} + \cdots \right) \\ &= \theta_1 + \frac{\theta_1 r^2}{2\theta_2^2} - \frac{\theta_1 r^3}{6\theta_2^3} + \cdots\end{aligned}$$

Thus the first odd power term is $-\frac{\theta_1 r^3}{3\theta_2^3}$, so the parameterization of this model should be $\phi_1 = \frac{\theta_1}{\theta_2^3}$ and the other $\phi_2 = \frac{1}{\theta_2}$.

(c) Considering the $\theta_1 e^{-r/\theta_2} (1 + r/\theta_2)$ model, we reparameterize it in terms of ϕ_1 and ϕ_2 as follows:

$$\theta_1 e^{-r/\theta_2} (1 + r/\theta_2) = \frac{\phi_1}{\phi_2^3} e^{-r\phi_2} (1 + r\phi_2).$$

We can also observe the Taylor expansion of this function as follows:

$$\begin{aligned}\frac{\phi_1}{\phi_2^3} e^{-r\phi_2} (1 + r\phi_2) &= \frac{\phi_1}{\phi_2^3} \left(1 - r\phi_2 + \frac{r^2\phi_2^2}{2} - \frac{r^3\phi_2^3}{6} + \cdots \right) (1 + r\phi_2) \\ &= \frac{\phi_1}{\phi_2^3} \left(1 - \frac{r^2\phi_2^2}{2} + \frac{r^3\phi_2^3}{3} + \cdots \right) \\ &= \frac{\phi_1}{\phi_2^3} - \frac{\phi_1 r^2}{2\phi_2} + \frac{\phi_1 r^3}{3} + \cdots.\end{aligned}$$

For our data $Y \sim N(\beta \mathbf{1}, \Sigma(\phi_1, \phi_2))$, where β is an unknown mean, we can use contrasts to get the REML estimates of ϕ_2 . Define $Z = C^\top Y$ where C is a contrast matrix such that $C^\top \mathbf{1} = 0$ and $C^\top C = I$. Then we have that $Z \sim N(0, C^\top \Sigma(\phi_1, \phi_2) C)$. For the REML log-likelihood (up to an additive constant) we can use the following formula:

$$\ell_R(\phi_1, \phi_2) = -\frac{1}{2} \log |C^\top \Sigma(\phi_1, \phi_2) C| - \frac{1}{2} Z^\top (C^\top \Sigma(\phi_1, \phi_2) C)^{-1} Z.$$

Next, we observe $C^\top \Sigma(\phi_1, \phi_2) C$ as $\phi_2 \rightarrow 0$. Let us first define our covariance matrix $\Sigma(\phi_1, \phi_2)$ as follows:

$$\Sigma(\phi_1, \phi_2) = (K(r_{ij}))_{i,j}.$$

For convenience we define $(R_{\phi_2})_{ij} = e^{-r_{ij}\phi_2} (1 + r_{ij}\phi_2)$, then we can write $\Sigma(\phi_1, \phi_2) = \frac{\phi_1}{\phi_2^3} R_{\phi_2}$. Therefore the contrast covariance matrix can be written as

$$C^\top \Sigma(\phi_1, \phi_2) C = \frac{\phi_1}{\phi_2^3} C^\top R_{\phi_2} C.$$

Now observing R_{ϕ_2} as a matrix Taylor expansion, we have that

$$R_{\phi_2} = \mathbf{1}\mathbf{1}^\top - \frac{\phi_2^2}{2}R^{(2)} + \frac{\phi_2^3}{3}R^{(3)} + \dots,$$

where $R^{(2)}$ and $R^{(3)}$ are the matrices of the second and third powers of the distances between the points. Thus we have that

$$C^\top R_{\phi_2} C = C^\top \mathbf{1}\mathbf{1}^\top C - \frac{\phi_2^2}{2}C^\top R^{(2)}C + \frac{\phi_2^3}{3}C^\top R^{(3)}C + \dots.$$

Since $C^\top \mathbf{1} = 0$, we lose the first term and we have that

$$C^\top R_{\phi_2} C = -\frac{\phi_2^2}{2}C^\top R^{(2)}C + \frac{\phi_2^3}{3}C^\top R^{(3)}C + \dots.$$

Thus we have that

$$C^\top \Sigma(\phi_1, \phi_2)C = -\frac{\phi_1}{2\phi_2}C^\top R^{(2)}C + \frac{\phi_1}{3}C^\top R^{(3)}C + \dots.$$

Next we observe the leading term to understand the behavior of the REML estimate as $\phi_2 \rightarrow 0$. Let the locations of the points be given by the rows of a matrix $X \in \mathbb{R}^{m \times k}$. If we consider that $r_{ij} = \|x_i - x_j\|$ then $R_{ij}^{(2)} = r_{ij}^2$. By the identity $r_{ij}^2 = \|x_i\|^2 + \|x_j\|^2 - 2x_i^\top x_j$, we can write $R^{(2)}$ as follows, with s the m -vector of squared norms of the rows of X :

$$R^{(2)} = \mathbf{1}s^\top + s\mathbf{1}^\top - 2XX^\top.$$

When we conjugate this with C , we have that

$$C^\top R^{(2)}C = -2C^\top XX^\top C.$$

Let us define $B = X^\top C$, then we have that

$$C^\top R^{(2)}C = -2B^\top B.$$

We know that $B^\top B$ is positive semidefinite and has rank at most k . Thus $C^\top R^{(2)}C$ is negative semidefinite and has rank at most k . We can further say that $C^\top R^{(2)}C$ has exactly rank k since C has full column rank $m - 1$ and X has full column rank k . Thus all k nonzero eigenvalues of $C^\top R^{(2)}C$ are negative.

If $\lambda_1, \dots, \lambda_k$ are the nonzero eigenvalues of $C^\top R^{(2)}C$, then for small ϕ_2

$$C^\top \Sigma(\phi_1, \phi_2)C \approx -\frac{\phi_1}{2\phi_2}C^\top R^{(2)}C,$$

so the k nonzero eigenvalues of $C^\top \Sigma(\phi_1, \phi_2)C$ are approximately given by $\sim c_i \frac{\phi_1}{\phi_2}$ where $c_i = -\lambda_i/2 > 0$. Hence

$$\log |C^\top \Sigma(\phi_1, \phi_2)C| = \sum_{i=1}^{m-1} \log \lambda_i(\phi_2) \sim k \log \left(\frac{\phi_1}{\phi_2} \right) + O(1),$$

and therefore

$$-\frac{1}{2} \log |C^\top \Sigma(\phi_1, \phi_2) C| \sim -\frac{k}{2} \log \left(\frac{\phi_1}{\phi_2} \right) + O(1).$$

For the quadratic term in ℓ_R , diagonalize $C^\top \Sigma(\phi_1, \phi_2) C$ and write Z in the corresponding eigenbasis. Then

$$-\frac{1}{2} Z^\top (C^\top \Sigma(\phi_1, \phi_2) C)^{-1} Z = -\frac{1}{2} \sum_{i=1}^{m-1} \frac{\tilde{z}_i^2}{\lambda_i(\phi_2)}.$$

For the k large eigenvalues we have $\lambda_i(\phi_2) \sim c_i \phi_1 \phi_2^{-1}$, so $1/\lambda_i(\phi_2) = O(\phi_2) \rightarrow 0$ as $\phi_2 \rightarrow 0$, and these contributions vanish. The remaining $m - k - 1$ eigenvalues stay bounded away from zero and infinity, so the corresponding terms converge to a finite limit. Thus the quadratic term remains finite as $\phi_2 \rightarrow 0$.

Combining these results, as $\phi_2 \rightarrow 0$ the determinant term in $\ell_R(\phi_1, \phi_2)$ behaves like $-\frac{k}{2} \log(\phi_1/\phi_2) \rightarrow -\infty$, while the quadratic term stays finite. Hence, for this parametrization and an unknown constant mean, the restricted log-likelihood does *not* tend to a finite value as the inverse range parameter ϕ_2 tends to 0.

2. 6.13

Solution:

$$K(r) = \frac{\phi_1}{\phi_2} \exp(-\phi_2 r),$$

To study how often the REML estimate of ϕ_2 hits the boundary at 0, I ran the following simulation. I fixed $\phi_1 = 1$ and considered sample sizes

$$m \in \{5, 10, 50, 100, 200\}$$

and true inverse ranges

$$\phi_2 \in \{0.2, 0.5, 1, 2, 5\}.$$

For each pair (m, ϕ_2) I repeated the steps below $N = 200$ times:

1. Generate m locations independently from $\text{Unif}([0, 1]^2)$ in the unit square.
2. Form the $m \times m$ covariance matrix $\Sigma(\phi_1, \phi_2)$ with entries $\Sigma_{ij} = K(r_{ij}) = (\phi_1/\phi_2) \exp(-\phi_2 r_{ij})$.
3. Simulate one realization $Y \sim N(\beta \mathbf{1}, \Sigma(\phi_1, \phi_2))$ with $\beta = 0$.

4. Fit the same model by restricted maximum likelihood, using `optimize` to maximize the restricted log-likelihood over $\phi_2 \geq 0$ while profiling out ϕ_1 and the mean.
5. Record whether the REML estimate satisfies $\hat{\phi}_2^{(R)} = 0$. In the code I treat $\hat{\phi}_2^{(R)} < 10^{-4}$ as 0.

For each (m, ϕ_2) I then estimated $\hat{p}(m, \phi_2) = \frac{1}{N} \sum_{r=1}^N \mathbf{1}\{\hat{\phi}_{2,r}^{(R)} = 0\}$. The plot below shows $\hat{p}(m, \phi_2)$ as a function of m (on a log scale) for each fixed value of ϕ_2 .

Numerically, for $\phi_2 = 0.2$ I obtained

$$\hat{p}(5, 0.2) = 0.35, \quad \hat{p}(10, 0.2) = 0.355, \quad \hat{p}(50, 0.2) = 0.295, \quad \hat{p}(100, 0.2) = 0.32, \quad \hat{p}(200, 0.2) = 0.34,$$

so the probability that the REML estimate hits the boundary stays around 0.3 and does not seem to decrease as m increases. For $\phi_2 = 0.5$ the estimated probabilities lie between about 0.24 and 0.45 over this range of m , again clearly bounded away from 0. For $\phi_2 = 1$ the probability decreases from about 0.40 at $m = 5$ to about 0.12 at $m = 200$. For larger inverse ranges, $\phi_2 = 2$ and $\phi_2 = 5$, the probability becomes quite small: for $\phi_2 = 2$ it drops to roughly 0.02 by $m = 200$, and for $\phi_2 = 5$ it is essentially 0 once $m \geq 50$.

Overall, these simulations suggest that for small inverse ranges (i.e., very smooth, strongly correlated fields) the probability that $\hat{\phi}_2^{(R)} = 0$ does not tend to 0 as m increases. Instead, for values like $\phi_2 = 0.2$ and $\phi_2 = 0.5$, $\hat{p}(m, \phi_2)$ appears to stabilize around a positive level. For larger inverse ranges, such as $\phi_2 = 2$ or $\phi_2 = 5$, the probability does decrease toward 0 as m grows. This is consistent with the discussion in Section 6.3 and the restricted log-likelihood plots in Figure 6.11: because the restricted log-likelihood has a finite limit as $\phi_2 \downarrow 0$ for this parametrization and an unknown constant mean, there can be a non-negligible chance that the REML maximizer over $\phi_2 \geq 0$ occurs at the boundary, especially when the true process has a very large spatial range.

Note that I got help for the code and simulation approach from George and Luke.

Probability that REML estimate of inverse range hits 0

